SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.*

QUESTIONS: 1 **Total Marks:100**

Annexure A

Technical Presentation

Code of Conduct:

I B. Srimathi (192511448) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Srimathi B



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

REAL TIME OPERATING SYSTEM

CSA0414-Operating system

Guided By

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Presented By

SRIMATHI B
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Introduction to Real Time Operating System

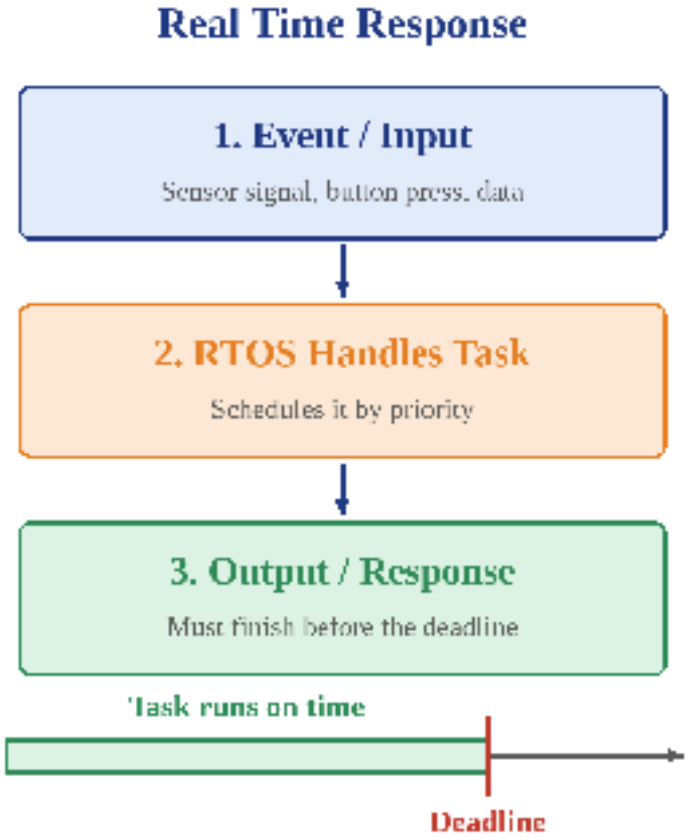
Definition: An operating system that gives the correct output within a fixed time limit.

Key idea: A right answer is useful only if it comes at the right time.

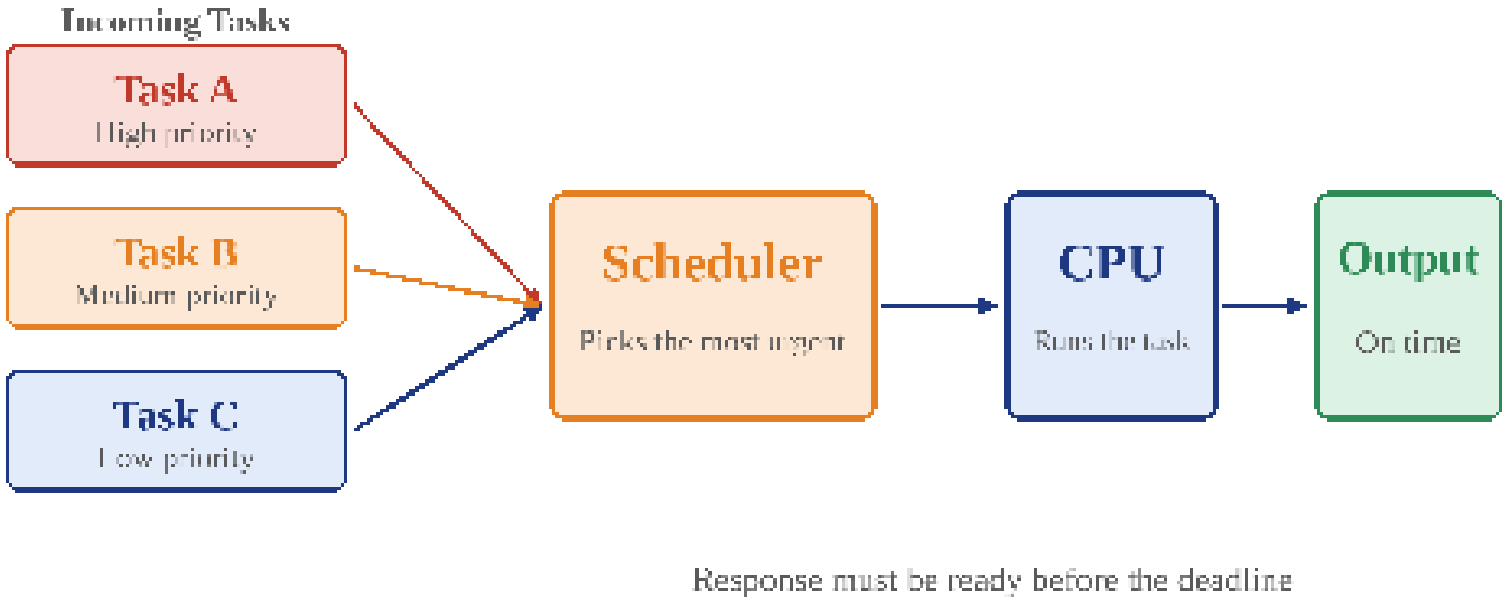
Deadline: Every task has a deadline that the system must meet.

Priority: Urgent tasks run first and can pause less important ones.

Examples: Car airbags, pacemakers, robots and flight control.



How a Real Time Operating System Works



Step 1: A task arrives from an input such as a sensor or a button.

Step 2: The scheduler checks the priority and deadline of every waiting task.

Step 3: The most urgent task gets the CPU, and a lower task is paused if needed.

Step 4: The output is sent on time, and the next task is picked.

Types and Applications

Types

Hard RTOS: missing a deadline is a failure, as in airbags and pacemakers.

Soft RTOS: a late response lowers quality only, as in video streaming.

Firm RTOS: a late result is useless but not dangerous, as in some factory lines.

Applications

Vehicles: engine control, anti-lock brakes and airbags.

Medical and aerospace: pacemakers, ventilators and flight control.

Robots and phones: factory robots, drones and mobile network control.

Thank you!!

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